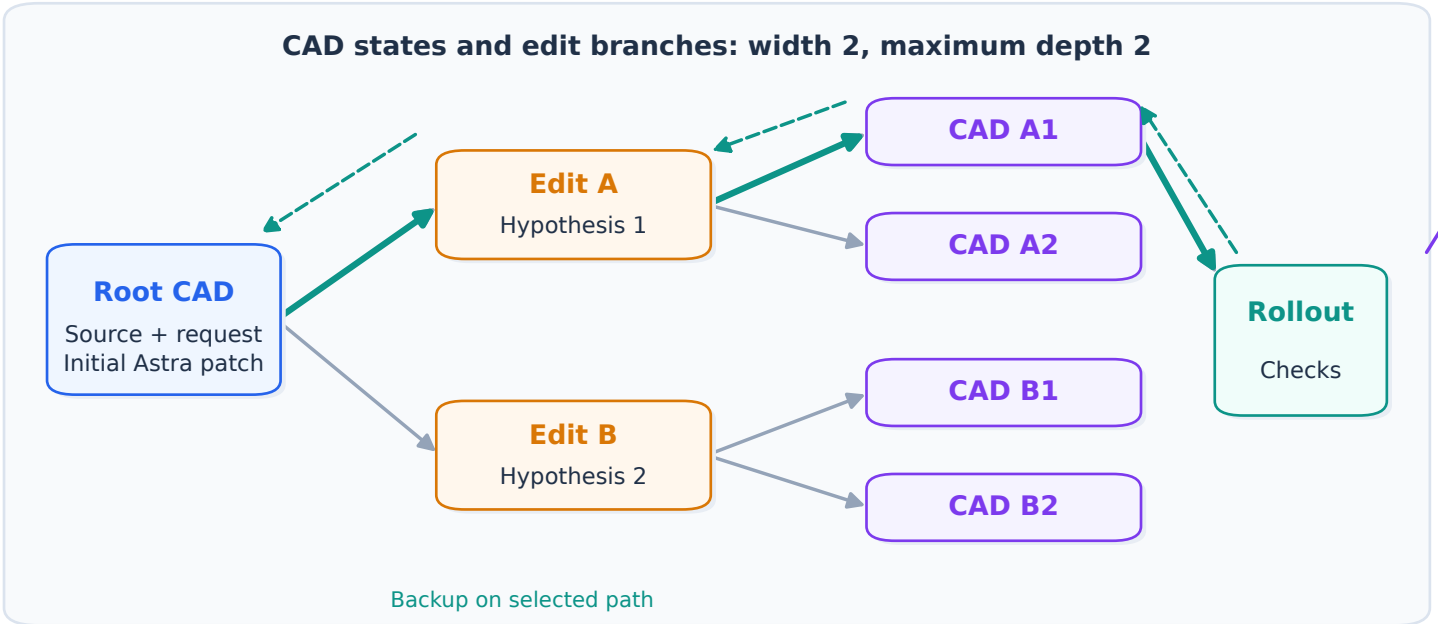
Monte Carlo Tree Search over CAD edit thoughts

COMPLETED PILOT



Target-free verification

Execute / numeric checks
Instruction consistency
Frozen GRPO verifier

Gate blocked every candidate

Max calibrated p: 0.8997 < 0.9016
All initial answers retained

A proxy score is not proof
of reference agreement

1 SELECT

Value + exploration

2 EXPAND

Sample Astra patches

3 ROLLOUT

Execute + proxy score

4 BACK UP

Update path Q and N

Repeat until fixed API / token / tool / depth budget is exhausted; reference geometry is used only for final offline scoring.